Problem for R1 Batch (Enrolment Numbers BT21CSE001 to BT20CSE025):

To conduct any examination, an eligible candidate database should be maintained by the organization. For example, consider the semester examination of an education institute. Let for the current semester, the examination will be conducted for 2nd, 4th, 6th and 8th semester students for all departments. There are five departments namely, Computer Sc. & Engg., Electronics & Comm., Electrical, Mechanical and Civil. The corresponding department codes are CS, EC, EE, ME and CV respectively. Also consider that every student has registered for 5 different courses belonging to his/her department.

Create a database of ***general student records*** comprises of the following information:

- Name of student
- Department code
- Year/Sem
- Roll number (unique key across dept. and year)
- Number of classes conducted for each of the subjects registered.
- Number of classes attended by the student for each of the subjects registered.

To track the attendance eligibility, only one subject (for which maximum number of classes were conducted) is considered for every semester and every department, and is denoted as MCC. The percentage of attendance is computed as below-

The attendance of the student, who has maximum number of classes attended for MCC, is considered as 100%. With respect to this, % of attendance for other students as required is to be computed using his/her attendance for MCC.

Create four attendance record databases, each database corresponds to one semester, to maintain the attendance of all student in the particular semester irrespective of their departments. The following information are to be kept in the attendance database:

- Roll number
- Department
- Attendance : Number of class attended for MCC and % attendance

Create fee status records database to maintain the record of the fee status for all students. The database contains the below information:

- Roll number
- Fee status (pending/clear)

Create an applicant records database to track the record of the students who have applied for the examination. The database stores the following fields:

- Name of the student
- Roll number
- A/NA (A stands for applicant, NA stands for non-applicant)

Create the ***general student records*** database by reading the data (either from the users or through file reading operations).

Write efficient C code to accomplish the following operations.

1. Sort the ***general student record*** database based first on semester, and then followed by department-name, and then the roll number.
2. Display ***year wise*** and then the ***department-wise*** names of the students who did not apply for the examination.
3. A student is eligible to sit for the semester examination if his/her attendance is more than 75% for MCC, fees status is clear and he/she applied for the examination. Create a list of all eligible students for the semester examination. The list of students is to be based first on semester, and then followed by department-name, and then the roll number.
4. Create a list of students having attendance <=75%. based first on semester, and then followed by department-name, and then the roll number.
5. Delete a student record. Input is roll number.
6. Print the name of the students whose attendance is >75% for the respective MCC but their fee status is pending.
7. Create a list of defaulter students. A student is called as defaulter either his/her attendance for MCC is <=75% or his/her fee status is pending. Find the name of the department in which maximum number of defaulters belong to.
8. Range-search students based on the roll number. Inputs are two roll numbers enrol1 and enrol2, and the output is the list of students having roll numbers *larger* than *enrol1* and *less* than *enrol2*.

Note: The information regarding the fields corresponding to each database (student records, attendance records, fee status records and applicants' records) should be maintained strictly. It is desirable to take the input through file handling. The program should be executed by taking the records of at least 100 students as input (minimum five students for every year and department).